Irrigation Efficiency and Productivity in the Ogallala Aquifer Region: The Role of Weather

Abstract

This article examines how accounting for local weather conditions changes measures of both irrigation-oriented and output-oriented efficiency and productivity of irrigation practices in the High Plains Aquifer (HPA) region. Using three-stage Data Envelopment Analysis and the Global Malmquist Index, we analyze county-level data from five agricultural census years (1997–2017) to compare results with and without weather variables (precipitation and extreme degree days). Including weather raises the average irrigation-oriented efficiency score by 21.4% (from 0.56 to 0.68) and the output-oriented score by 6.2% (from 0.81 to 0.86). The share of counties on the irrigation best-practice frontier doubles from 13.9% to 28.0%. Irrigation productivity growth over the full period increases from 120% to 146% when weather is accounted for, with efficiency change nearly three times higher (96% vs. 34%). The largest adjustments are concentrated in southwestern Kansas and the Texas–Oklahoma Panhandle. These results demonstrate that omitting weather variables substantially understates both efficiency levels and productivity growth while misattributing environmental adaptation to other sources of change.

Key words: Data Envelopment Analysis, Global Malmquist Index, High Plains Aquifer, Irrigation Efficiency, Ogallala Aquifer, Productivity, Weather

JEL codes: Q15, Q25, D24, C14

The High Plains Aquifer (HPA), also known as the Ogallala Aquifer (figure 1), provides about 30% of the nation's irrigation water (Steward et al. 2013), increasing the value of crop production by over $3 billion annually (Garcia-Suarez, Fulginiti, and Perrin 2019). The fact that it has been depleted by an average of about 16 feet since predevelopment (McGuire 2017) raises concern about the allocation efficiency and the productivity of these waters. Here we address those issues by measuring the efficiency and productivity path of irrigation across the HPA, giving special attention to the issue of distinguishing the productivity and efficiency of discretionary choices (irrigation, fertilizer, chemicals), as distinct from the effects of weather that varies across space and time.

No previous studies of the HPA have attempted to distinguish the efficiency and productivity impacts of weather as opposed to discretionary choices by producers. Lilienfield and Asmild (2007) analyzed water-use efficiency in Western Kansas but did not attempt to purge the effects of weather from their measure. Yet, rainfall and temperature vary sharply across the High Plains, which can potentially have large effects on measures of how efficiently farmers irrigate. Mean annual precipitation ranges from roughly 32 in yr^{-1} ($\approx$ 800 mm) on the aquifer's eastern edge to about 14 in yr^{-1} ($\approx$ 350 mm) in the west (McGuire et al. 2003; Anandhi et al. 2023). Regarding temperatures, average spring highs decline from $\approx$ 14°C in the south to $\approx$ 3°C in the north, and summer highs from $\approx$ 27°C to $\approx$ 18°C (Anandhi et al. 2023). Together with differences in soils and crop mixes, this weather heterogeneity produces wide spatial gaps in pumping volumes, the choice of irrigation technology (e.g., center-pivot versus drip), and the energy cost per acre-inch applied (Colaizzi et al. 2009).

This article addresses this gap by systematically comparing irrigation efficiency and productivity measures calculated with and without weather variables. Our objectives are to: (a) determine how including weather variables (precipitation and extreme degree days) affects measured irrigation-oriented and output-oriented technical efficiency scores across HPA counties, with emphasis on irrigation given the aquifer's depletion concerns; and (b) assess how weather variables affect measured irrigation productivity change over time.

Our primary finding is that weather effects result in substantial underestimates of measured irrigation efficiency and productivity. Irrigation efficiency scores increase by 21.4% on average when weather is included (from 0.56 to 0.68), while irrigation productivity growth over the full period increases from 120% to 146%. These adjustments are largest in southwestern Kansas and the Texas–Oklahoma Panhandle.

A limitation of our study is that we examine irrigation heterogeneity through technical efficiency only on the extensive margin (share of land irrigated), as data on the intensive margin (amount of water per acre) is unavailable. This constraint is relevant since some papers in the literature of irrigation, including Hendricks and Peterson (2012), show that irrigation adjustments occur primarily through the intensive margin, suggesting our analysis may not capture all dimensions of irrigation heterogeneity in the region.

The remainder of this article is organized as follows: the next section presents the theoretical framework and methods, followed by the data description, results, and conclusions.

Theory and Method

Theoretical Model

Consider an agricultural production system where discretionary input bundles are denoted by $\boldsymbol{x}$. Non-discretionary input bundles (i.e., those not controlled by the producer) are denoted by $\boldsymbol{z}$. Each discretionary input bundle is defined on a nonnegative, S-dimensional real space, represented as $\boldsymbol{x} \in \mathbb{R}_+^S$. Non-discretionary input bundles are defined on a nonnegative, R-dimensional real space, denoted as $\boldsymbol{z} \in \mathbb{R}_+^R$. The output bundles of this system are represented by $\boldsymbol{y}$, defined on a nonnegative, M-dimensional real space, expressed by $\boldsymbol{y} \in \mathbb{R}_+^M$.

In this article, we use a single output ($M = 1$), three discretionary inputs ($S = 3$), and two non-discretionary inputs ($R = 2$). Output is crop sales per acre, and discretionary inputs are share of cropland irrigated, fertilizer per acre, and chemicals per acre. Precipitation and extreme degree days (above 30 Celsius degrees) are considered non-discretionary inputs. More information about the data is presented in the Data and Estimation section.

We define an input requirement set $L(y)$ as follows:

$$L(y) = \left\{ \boldsymbol{x} \in \mathbb{R}_{+}^{S} : (\boldsymbol{x}, y) \in T(\boldsymbol{z}) \right\} \tag{1}$$

where $T(\boldsymbol{z})$ represents the production possibilities set for given weather inputs $\boldsymbol{z}$. We will refer to $T(\boldsymbol{z})$ as T. T is the set of all technically feasible input and output combinations (Chambers 1988). We assume that T is a nonempty, closed and convex set, and bounded from above for every finite $\boldsymbol{x}$. In addition, we assume $\boldsymbol{x}$ and $\boldsymbol{y}$ are freely disposable in T, and output cannot be produced when the input bundle equals zero.

$L(y)$ is convex, and nonempty for at least one finite output level y. In addition, if $y' \geq y$, then for an input bundle $\boldsymbol{x}$, $L(y') \subseteq L(y)$. Finally, $L(y)$ is characterized by diminishing marginal rate of technical substitution. Expression (1) indicates all possible input bundles that can be used to produce the level of output y, given the state of the technology.

In figure 2, we represent the output set $P(\boldsymbol{x})$ having only one of the inputs (irrigation) on the horizontal axis. By doing so, we are assuming that all other inputs are fixed at a given level. The set $P(\boldsymbol{x})$ is composed of the gray area and the black concave line in figure 2. We call this concave line the production function (or production frontier) which can be represented as:

$$f(\boldsymbol{x} \mid \boldsymbol{z}) = \max \{ y : \boldsymbol{x} \in L(y) \} \tag{2}$$

Therefore, the production function represents the maximum output that can be produced, given irrigation and given all other inputs. Note that no combination of inputs and outputs is feasible above or on the left of the production frontier, since it is infeasible, given the production technology.

If a Decision Making Unit (DMU) is on the frontier, it is considered efficient, and if it lies below the production frontier, it is considered technically inefficient. A technically inefficient DMU is not maximizing production, given its input usage. An inefficient DMU should be able to produce more output using the same inputs, or use less inputs to produce the same amount of output. Because weather varies substantially across the High Plains (precipitation ranges from 14 to 32

inches annually; summer temperatures vary by ~10°C), comparing counties without accounting for these differences may conflate environmental constraints with inefficiency.

In figure 2, it is straightforward to note that DMU B and C are efficient, and DMU A is inefficient. DMU A would be efficient by increasing the amount of y while using the same amount of irrigation (i.e., by imitating DMU B). This approach will be the one used to measure output-oriented technical efficiency. DMU A would also be efficient if it uses less inputs while keeping the same amount of production to achieve the efficient bundle used by DMU C. We use this approach to measure irrigation-oriented technical efficiency.

Empirical Model

We calculate two types of technical efficiency: output-oriented (how much could output increase given inputs?) and irrigation-oriented (how much could irrigation decrease given output and other inputs?). Both employ Ruggiero's (1998) three-stage Data Envelopment Analysis approach, which accounts for non-discretionary inputs (variables beyond farmers' control). In our context, these non-discretionary inputs are weather variables: precipitation and extreme degree days.

Throughout this section, we use θ for the output-oriented LP decision variable and φ for the irrigation-oriented LP decision variable. Parenthesized superscripts denote the stage of the DEA procedure: (1) for the first stage (without weather controls), (3) for the third stage (with weather adjustment), and (G) for the global technology used in the Malmquist index. An asterisk denotes the optimal value from the linear program.

We begin with output-oriented efficiency, which follows the standard DEA approach. However, given aquifer depletion concerns, our primary interest is irrigation-oriented efficiency, which requires the sub-vector DEA approach to isolate irrigation's efficiency while holding other inputs constant.

Output-Oriented Technical Efficiency without and with Weather Impacts

Output-oriented efficiency measures maximum output achievable given observed inputs and weather conditions. This provides a baseline for comparison with our primary measure of interest: irrigation

efficiency.

Stage 1: Standard Output-Oriented DEA without Weather Controls

$$
\begin{aligned}
\max_{\theta^{(1)},\boldsymbol{\lambda}} \quad & \theta^{(1)} \\
\text{s.t.} \quad & \sum_{k=1}^{K} \lambda_k \, x_{nk} \leq x_{ni}, \quad n = 1, \ldots, N \\
& \theta^{(1)} \, y_i \leq \sum_{k=1}^{K} \lambda_k \, y_k \\
& \sum_{k=1}^{K} \lambda_k = 1 \\
& \lambda_k \geq 0, \quad k = 1, \ldots, K
\end{aligned}
\tag{3}
$$

where the superscript i denotes the DMU under analysis and k indexes other DMUs. Also, y_k is the observed level of output produced by the kth DMU, x_{nk} is the observed level of the nth input used by the kth DMU, and K depicts the total number of DMUs. Finally, λ's are used to construct the production frontier as a convex combination of the Decision-Making Units (DMUs) in the reference set. These weights also indicate which observations serve as benchmarks for the specific DMU under analysis, helping to determine their relative efficiency. The λ values are calculated as part of the linear programming problem.

The scalar $\theta^{(1)*}$, which is the maximum expansion factor, represents the factor by which the county's observed output could be proportionally increased while using the same inputs. If $\theta^{(1)*} = 1.25$, for example, the county could theoretically increase crop sales by 25% with its current input levels by adopting the practices of benchmark counties on the efficiency frontier.

However, for consistency with our irrigation efficiency measure (discussed below), we report the inverse, $1/\theta^{(1)*}$. This inverse transformation provides several advantages. First, the inverse score is bounded between 0 and 1, where 1 indicates perfect efficiency (the county is on the frontier) and lower values indicate inefficiency. This is more intuitive than $\theta^{(1)*}$, where higher values indicate

worse performance. Second, the inverse allows direct comparison with irrigation efficiency scores, which are naturally bounded $[0, 1]$.

Stage 2: Measuring Weather's Impact on Output Efficiency

The first-stage scores do not account for weather differences across counties. A county facing drought and extreme heat may appear inefficient when compared to counties with favorable growing conditions, even if both are managing their resources optimally given their respective environments. To isolate weather's effect, we regress the first-stage efficiency scores on weather variables using a logistic transformation:

$$\text{logit}\left(1/\theta_i^{(1)*}\right) = \beta_1 \, EDD_i + \beta_2 \, Precip_i + \gamma_i + \delta_t + \varepsilon_{it} \tag{4}$$

The logistic transformation, $\text{logit}(p) = \ln[p/(1-p)]$, is necessary because the inverse efficiency scores are bounded between 0 and 1. Therefore, standard linear regression would be inappropriate for such bounded dependent variables. The independent variables are extreme degree days (*EDD*), defined as cumulative exposure to temperatures above 30°C during the growing season, and cumulative precipitation during the same period. We include county fixed effects (γ_i) to control for time-invariant county characteristics such as soil quality, and year fixed effects (δ_t) to account for region-wide shocks.

The estimated coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ reveal how weather affects measured efficiency. A negative coefficient on *EDD* ($\hat{\beta}_1 < 0$) would indicate that extreme heat reduces efficiency scores, while a positive coefficient on precipitation ($\hat{\beta}_2 > 0$) would suggest that adequate rainfall improves scores. Using these estimated coefficients, we construct an overall environmental harshness index for each county-year observation:

$$E_{it}^{y} = \hat{\beta}_1 \, EDD_{it} + \hat{\beta}_2 \, Precip_{it} \tag{5}$$

Lower values of this index indicate harsher environmental conditions. For example, a county-year

with high degree days and low precipitation would have a low (more negative) environmental index, reflecting the challenging growing conditions that year.

Stage 3: Recalculating Output Efficiency with Weather Adjustment

In the third stage, we recalculate efficiency scores while explicitly accounting for the environmental conditions each county faces. We solve the same optimization problem as Stage 1, but now considering the environment to which DMUs are exposed:

$$
\begin{aligned}
\max_{\theta^{(3)},\boldsymbol{\lambda}} \quad & \theta^{(3)} \\
\text{s.t.} \quad & \sum_{k=1}^{K} \lambda_k \, x_{nk} \leq x_{ni}, \quad n = 1, \ldots, N \\
& \theta^{(3)} \, y_i \leq \sum_{k=1}^{K} \lambda_k \, y_k \\
& \sum_{k=1}^{K} \lambda_k = 1 \\
& \lambda_k = 0 \quad \text{if } E_k^y > E_i^y \\
& \lambda_k \geq 0, \quad k = 1, \ldots, K
\end{aligned}
\tag{6}
$$

Compared with equation (3), equation (6) adds a new constraint related to the overall index of environmental harshness (E^y). This constraint imposes that a DMU with better environment cannot be a benchmark for a DMU with worse environment. Overall, it prevents DMUs from having a very low efficiency score when they face bad weather.

This weather adjustment has important implications. Consider two counties: County A faces severe drought (low precipitation, high extreme heat), while County B enjoys favorable weather. In Stage 1, County A might appear highly inefficient because it is compared to all counties, including County B. In Stage 3, County A is compared only to counties facing similar or worse weather, preventing the unfair comparison. As a result, County A's efficiency score will typically increase (or remain the same) in Stage 3 relative to Stage 1, reflecting that some of its apparent inefficiency was due to environmental constraints rather than poor management.

We denote the final output-oriented efficiency score as $1/\theta_i^{(3)*}$. If $1/\theta_i^{(3)*} = 1$, the county-year observation is technically efficient. If $1/\theta_i^{(3)*}$ is between 0 and 1, the county-year observation is inside the production frontier, and therefore is technically inefficient. To assess weather's impact on measured efficiency, we compare the Stage 3 scores ($1/\theta^{(3)*}$, with weather) to Stage 1 scores ($1/\theta^{(1)*}$, without weather) throughout our analysis.

Irrigation-Oriented Technical Efficiency without and with Weather Impacts

We are particularly interested in calculating the efficiency measure in the use of irrigation alone. Because the efficiency score is calculated for only one of the inputs (and not for the entire input vector), we employ the sub-vector variation of the input-oriented DEA to calculate the irrigation-oriented technical efficiency score (Färe, Grosskopf, and Lovell 1994). As with the calculation of output-oriented technical efficiency scores, we employ a three-step approach for the irrigation-oriented technical efficiency scores in order to control for weather.

Stage 1: Sub-Vector Irrigation-Oriented DEA without Weather Controls

Denote w_k as the observed share of land irrigated used by the kth DMU. We can calculate the first-step irrigation-oriented efficiency score for a specific DMU (i) using the following sub-vector DEA:

$$
\begin{aligned}
\min_{\varphi^{(1)},\boldsymbol{\lambda}} \quad & \varphi^{(1)} \\
\text{s.t.} \quad & \varphi^{(1)}\, w_i \geq \sum_{k=1}^{K} \lambda_k\, w_k \\
& \sum_{k=1}^{K} \lambda_k\, x_{nk} \leq x_{ni}, \quad n = 1, \ldots, S \\
& \sum_{k=1}^{K} \lambda_k\, y_k \geq y_i \\
& \sum_{k=1}^{K} \lambda_k = 1 \\
& \lambda_k \geq 0, \quad k = 1, \ldots, K
\end{aligned}
\tag{7}
$$

The main difference between equation (7) and the first stage of the calculation of the output-oriented efficiency in equation (3) is that we minimize the scalar $\varphi^{(1)}$ rather than maximizing it. In the input-oriented measure, we seek the smallest scalar that, when multiplied by the observed level of irrigation (w_i), still allows the county to achieve its observed output level. $\varphi^{(1)*}$ is bounded between 0 and 1, with county-year observations having values equal to 1 being technically efficient in irrigation use, while county-year observations with values between 0 and 1 are considered technically inefficient in irrigation use.

Stage 2: Measuring Weather's Impact on Irrigation Efficiency

Counties facing drought and extreme heat may require more share of irrigated land to achieve the same output as counties with adequate rainfall and moderate temperatures. To capture this effect, we regress first-stage irrigation efficiency scores on weather variables:

$$\text{logit}\left(\varphi_i^{(1)*}\right) = \alpha_1 \, EDD_i + \alpha_2 \, Precip_i + \gamma_i + \delta_t + \varepsilon_{it} \tag{8}$$

The logistic transformation again accommodates the bounded nature of efficiency scores. We expect $\hat{\alpha}_1 < 0$ (extreme heat necessitates more irrigation, reducing efficiency scores) and $\hat{\alpha}_2 > 0$ (precipitation substitutes for irrigation, improving efficiency scores). Then, we construct the overall index of environmental harshness for each DMU—denoted by E^w:

$$E_{it}^w = \hat{\alpha}_1 \, EDD_{it} + \hat{\alpha}_2 \, Precip_{it} \tag{9}$$

Stage 3: Recalculating Irrigation Efficiency with Weather Adjustment

Similar to equation (6), we recalculate the irrigation-oriented technical efficiency score by taking into account the overall index of environmental harshness:

$$
\begin{aligned}
\min_{\varphi^{(3)},\boldsymbol{\lambda}} \quad & \varphi^{(3)} \\
\text{s.t.} \quad & \varphi^{(3)}\, w_i \geq \sum_{k=1}^{K} \lambda_k\, w_k \\
& \sum_{k=1}^{K} \lambda_k\, x_{nk} \leq x_{ni}, \quad n = 1, \ldots, S \\
& \sum_{k=1}^{K} \lambda_k\, y_k \geq y_i \\
& \sum_{k=1}^{K} \lambda_k = 1 \\
& \lambda_k = 0 \quad \text{if } E_k^w > E_i^w \\
& \lambda_k \geq 0, \quad k = 1, \ldots, K
\end{aligned}
\tag{10}
$$

The third-step irrigation technical efficiency score for a representative DMU (i) takes into account the environment harshness (E_i^w) of this DMU. As explained for the output-oriented case, for the calculation of the irrigation-oriented efficiency scores, the representative DMU is only compared to others that have equal or better weather (from the constraint $E_k^w \geq E_i^w$).

The irrigation-oriented efficiency score for a specific DMU, denoted as $\varphi_i^{(3)*}$, lies between 0 and 1, with a score of 1 indicating that the DMU under analysis is technically efficient in the use of irrigation. The irrigation-oriented technical efficiency score indicates the ratio between the potential and the observed levels of irrigation.

Unlike the output-oriented case, here we seek the smallest scalar that, when multiplied by the observed irrigation level, yields the minimum feasible irrigation. Multiplying $\varphi_i^{(3)*}$ by the observed irrigation level of a DMU calculates its technically efficient irrigation level, given its input use, output level, and environmental harshness.

For instance, if $\varphi_i^{(3)*} = 0.4$, DMU i's efficient or best practice level of irrigation is given by $0.4 \times w_i$. In this example, given the weather, output and other inputs, this DMU should be able to reduce the level of irrigation by 60% while keeping the same levels of production and use of other inputs.

In figure 2, the irrigation-oriented technically efficiency score illustrates the input usage ratio between observations C and A. By adjusting the irrigation level of observation A by its corresponding efficiency score $\varphi^{(3)*}$, the technically optimal level of irrigation of observation A would match the level used in best practice observation C.

Considering our focus on crop sales in the Ogallala aquifer region, we hypothesize that counties will exhibit higher output technical efficiency compared to irrigation use technical efficiency. This expectation is due to the common-pool resource nature of the aquifer. Since irrigation substantially boosts agricultural output and the only significant variable cost is associated with pumping, it is plausible that counties may operate with more irrigated land than is efficiently necessary.

When analyzing efficiency scores by year, we estimate equations (3) and (6) for output-oriented technical efficiency, and equations (7) and (10) for irrigation-oriented technical efficiency using data for each year. Then, we find the technical efficiency scores for each DMU, given the state of technology of that year.

Irrigation Productivity Change without and with Weather Impacts

In the previous section, we discussed how irrigation technical efficiency is calculated. While this measure helps us understand the heterogeneity among counties in each year of the sample, it does not reveal how the use of irrigation for output production is evolving over time. Specifically, technical efficiency measures do not tell us whether counties are using more or less irrigation in relation to other inputs (e.g., fertilizer and chemicals), given weather conditions throughout the sample period.

We are interested in understanding the evolution of productivity in the use of irrigated land. In order to calculate it, we use a Global Malmquist Index. This method allows analysis of productivity

change by taking into account technical change and efficiency change over time. The Global Malmquist Index for irrigation can be represented as follows (Pastor and Lovell 2005):

$$GMI^{t,t+1} = \frac{\varphi^{(G)}\left(\boldsymbol{x}^{t+1}, y^{t+1}, E^{w,t+1}\right)}{\varphi^{(G)}(\boldsymbol{x}^{t}, y^{t}, E^{w,t})} \tag{11}$$

where the vector of all inputs (including irrigation) is denoted by $\boldsymbol{x}$, the output is denoted by y, and the overall index of environmental harshness is denoted by E^{w}.

$GMI^{t,t+1}$ indicates the irrigation productivity change from period t to period $t+1$. If $GMI^{t,t+1} > 1$, water was more productively used in period $t+1$ than in period t. If $GMI^{t,t+1} < 1$, water was less productively used in period $t+1$ than in period t. Finally, if $GMI^{t,t+1} = 1$, then there was no difference in water productivity between period t and period $t+1$.

In equation (11), $\varphi^{(G)*}$ is the irrigation technical efficiency score using a global technology. The global technology is denoted by T^{G}. In panel data, T^{G} includes data for all counties in all years. Therefore, $\varphi^{(G)}$ is calculated using the same approach used to calculate the irrigation-oriented efficiency score $\varphi^{(3)*}$ in equation (10), but including pooled data to construct a meta-technology. Therefore, to calculate a global irrigation efficiency score for each DMU, instead of solving equations (7) and (10) for each year separately, we use pooled data—all years and counties together.

One of the virtues of the Global Malmquist Index is that it is circular (Pastor and Lovell 2005). For instance, assume we have data for 3 periods, t, $t+1$, and $t+2$. Therefore, the circularity property of the Global Malmquist Index indicates that:

$$GMI^{t,t+2} = GMI^{t,t+1} \times GMI^{t+1,t+2} \tag{12}$$

The Global Malmquist Index when using the values for outputs and inputs of periods t and $t+2$ (left-hand side) is equal to the cumulative productivity from period t to period $t+2$ (right-hand side). This is an important property of indexes as it implies that when comparing quantities across multiple time periods, the calculated quantity change is consistent regardless of the base

period, preventing any contradictions in rankings. Pastor and Lovell (2005) also show that we can decompose the measure of productivity change given by the Global Malmquist Index into efficiency change and technical change:

$$GMI^{t,t+1} = EC^{t,t+1} \times TC^{t,t+1} \tag{13}$$

where EC is efficiency change, and TC denotes technical change.

Figure 3 depicts an example of irrigation productivity change in output-irrigation space, in which we assume all other inputs are held fixed and we have data for two periods, t and $t + 1$.

The global technology, T^G, is the envelope of the technologies across time periods. In figure 3, at the first levels of irrigation, the global technology (red dashed line) is formed by the technology of period $t + 1$ (f^{t+1}, black solid line). At higher levels of irrigation, the global technology is determined by the frontier of period t (f^t, blue solid line).

Assume that a given county A generates a quantity $\bar{y}$ of crop sales in both periods (figure 3). In period t, county A uses w_A^t of irrigation, while in period $t + 1$ it uses w_A^{t+1}. Productivity change will be equal to the ratio between the technically efficient input bundle in period $t + 1$ and t with respect to the global technology frontier:

$$GMI_A^{t,t+1} = \frac{\varphi^{(G)}\left(\boldsymbol{x}_A^{t+1}, \bar{y}, E_A^{w,t+1}\right)}{\varphi^{(G)}\left(\boldsymbol{x}_A^{t}, \bar{y}, E_A^{w,t}\right)} = \frac{w_A^{G*,t+1}/w_A^{t+1}}{w_A^{G*,t}/w_A^{t}} \tag{14}$$

Because in this example $w_A^{G*,t+1}/w_A^{t+1} > w_A^{G*,t}/w_A^{t}$, irrigation productivity change is greater than 1. That means that from period t to period $t + 1$, county A became more productive in the use of irrigation. It is straightforward to realize that by looking at figure 3. In period $t + 1$, county A is using less irrigation than in period t, while producing the same level of output and using the same level of other inputs.

We can decompose the irrigation productivity change in equation (14) into efficiency change

and technical change, as in equation (13):

$$\underbrace{\frac{\varphi_A^{(G),t+1}}{\varphi_A^{(G),t}}}_{GMI} = \underbrace{\frac{\varphi_A^{(3),t+1}}{\varphi_A^{(3),t}}}_{EC} \times \underbrace{\frac{\varphi_A^{(G),t}/\varphi_A^{(3),t}}{\varphi_A^{(G),t+1}/\varphi_A^{(3),t+1}}}_{TC} \tag{15}$$

where $\varphi^{(3),t*}$ refers to the third-step irrigation technical efficiency score as explained in equation (10), $\varphi^{(G),t*}$ refers to the global irrigation efficiency score for period t, and similarly for period $t + 1$.

If, as in figure 3, the optimal level of irrigation in period t is greater than the optimal level of irrigation in period $t + 1$, $TC > 1$. Therefore, technology became more water-saving from period t to period $t + 1$. That is given by the shift to the left in the local frontier in $t + 1$. In figure 3, EC is also greater than one, since the distance between the observed level and the efficient level of irrigation in period t is greater than in period $t + 1$. It is also important to notice that the larger the horizontal distance of an observation to its frontier, the lower is the irrigation efficiency score.

Data and Estimation

The dataset is at the county-year level, and it comprises census years of 1997, 2002, 2007, 2012, and 2017. Only counties with at least 70% of area over the Ogallala aquifer were included in the dataset. In addition, we only keep counties with at least three observations in the final dataset.

Output is crop sales per harvested acre (in dollars of 2017). To express values in constant dollars, we deflate using the Producer Price Index by Commodity for Farm Products: Grains (WPU012), not seasonally adjusted, from the U.S. Bureau of Labor Statistics (2024).

Discretionary inputs are fertilizer per harvested acre (quantity index), chemicals per harvested acre (quantity index), and share of irrigated land (ratio). Non-discretionary inputs (weather environment variables) are accumulated precipitation (in centimeters), and degree days above 30 Celsius degrees. Weather variables are calculated for the growing season (May–September).

We use crop sales, rather than crop production, as the output variable due to the diversity of crops being planted in the Ogallala region. Crop production data could only be used if aggregated

into a single output, given that we lack information on fertilizer and chemical use by crop type.

Garcia-Suarez, Fulginiti, and Perrin (2019) use crop biomass as aggregate output. However, this approach can lead to issues because some crops may generate more value per unit of biomass than others. This difference in crop value is not captured by focusing solely on biomass.

One downside of using crop sales is that it does not directly reflect production due to the impact of storage. For example, if in a given year farmers store a higher proportion of their crop production, *ceteris paribus*, crop sales for that year will be lower, even if production remains unchanged. Conversely, farmers may sell previously stored production from earlier years, resulting in higher crop sales without a corresponding increase in production.

Crop sales, irrigated land, total harvested land, fertilizers and chemicals data were extracted from the Agricultural Censuses conducted by the National Agricultural Statistics Service of the U.S. Department of Agriculture (NASS, USDA). Data are available on the QUICKSTATS website. Accumulated precipitation and degree days above 30 Celsius degrees are calculated following Trindade (2011) and Trindade, Fulginiti, and Perrin (2023) using data from the United States Historical Climatology Network. Weather variables are calculated for the growing season (March–August).

Share of irrigated land (w) represents the fraction of irrigated land relative to total agricultural land in use, calculated as total irrigated harvested area divided by total area harvested.

The Agricultural Census reports county-level expenditures of fertilizers and chemicals, but not quantities. Therefore, as Garcia-Suarez, Fulginiti, and Perrin (2019), we convert fertilizer expenditures into implicit quantity indexes by dividing the expenditures by the fertilizer index, a national index reported by the ERS/USDA, and similarly for a chemicals quantity index. Finally, we divide the implicit quantities by the implicit quantity in Adams County, Nebraska, in 1960, to create the final indexes for chemicals and fertilizers.

Precipitation refers to the total amount of precipitation in the growing season, in centimeters. In this study, degree days above 30 Celsius degrees refers to the total number of full days of exposure to a temperature greater or equal to 30 Celsius degrees in the growing season.

Given that the DEA approach is sensitive to outliers, we first run an outlier test proposed by Wilson (2008). We run the test for each year separately, as the efficiency scores are calculated for each year separately. We include the output as well as the conventional inputs in the test—but not the weather variables, as they are not included in the DEA problem the same way as other inputs. After performing the test, three observations were excluded: Yuma county, CO, in 1997, York county, NE, in 2017 and Holt county, NE, in 2017. After excluding the outliers, our dataset is comprised of 518 observations for 118 counties.

Table 1 reports the descriptive statistics for the variables used in the article.

Crop sales averaged \$312.11 per acre (in 2017 dollars), with observations ranging from a minimum of \$28.13 to a maximum of \$775.07. The mean irrigation rate was 0.47, indicating that counties irrigated an average of 47% of their total planted area. Irrigation levels varied widely, from nearly no irrigation (3%) in some counties to almost full irrigation (96%) in others. Regarding other inputs use, counties on average applied 3.52 times the fertilizer and 12 times the chemicals compared to the levels observed in Adams, NE in 1960.

Regarding weather variables, we observe an average accumulated precipitation of 35.90 centimeters, but some observations are as low as 9.46 cm, while others are as high as 82 cm. Finally, observations averaged 8.97 degree days (of 24 hours each) of temperature above 30 degrees Celsius, also presenting high variability.

We calculate the output-oriented and irrigation-oriented technical efficiency scores without and with weather variables by solving the steps given by equations (3)–(6), and (7)–(10), respectively. The scores are calculated separately for each year; observations are compared with the technology obtained from the observations of the same year. We report the mean and standard deviation of the variables used in this article by year in table 2.

We calculate the Global Malmquist Productivity Index for irrigation by calculating global technical efficiency measures solving the linear programming problems of equations (7) and (10) for all counties and years (rather than one per year as in the previous subsection). We then use these global technical efficiency scores in equation (11) to obtain the Global Malmquist productivity

index and its components. To solve the linear programming problems, we use the `lp` package in the software R version 4.2.2.

Results

In this section we present the main findings of the article. First, we present the output-oriented and irrigation-oriented technical efficiency scores calculated using the results from the DEA programs of equations (6) and (10), respectively ($1/\theta^{(3)*}$ and $\varphi^{(3)*}$). We show the evolution of the scores over time, and the spatial distribution of the scores across counties over the High Plains Aquifer. Then, we proceed to calculate the irrigation productivity change from year to year, and the cumulated productivity using equation (11).

Output-Oriented and Irrigation-Oriented Efficiency Scores

To assess the role of weather variables in measuring irrigation efficiency, we present results from two specifications. Table 3 shows efficiency scores calculated with weather variables (precipitation and extreme degree days) included in the DEA model (third stage, equations 6 and 10). Table 4 presents results without weather controls (first stage, equations 3 and 7). Standard errors for the estimates are obtained from a 1,000-replication Simar and Wilson (2000) bootstrap that resamples counties with replacement and re-computes the DEA frontiers for each census year in every draw.

Our analysis reveals substantial variation in irrigation use across HPA counties. When weather variables are excluded from the model (table 4), the average irrigation-oriented efficiency score is 0.56, with only 13.9% of observations on the best-practice frontier. Including precipitation and extreme degree days (table 3) raises the mean score to 0.68, with 28.0% of observations now on the frontier.

This 21.4% increase in average irrigation efficiency demonstrates that weather conditions account for a substantial portion of the apparent heterogeneity in irrigation use across counties. Year-by-year comparisons show the weather adjustment raising irrigation efficiency scores by 6.8% in 1997, 12.3% in 2002, 27.8% in 2007, 15.1% in 2012, and 43.3% in 2017.

Even after accounting for weather, the average county uses approximately 47% more irrigated land than benchmark counties, given crop sales and other input use. However, important factors not included in our model (such as capital, labor, soil quality, managerial ability, local aquifer characteristics, and varying state regulations) likely explain much of this remaining variation. Therefore, our measures primarily capture heterogeneity across counties rather than strict inefficiency.

Turning briefly to output efficiency, the weather adjustment is more modest than for irrigation. Output-oriented efficiency scores show less variation than irrigation-oriented scores, both across counties and between the models with and without weather variables. The average output efficiency score increases from 0.81 without weather controls to 0.86 with them, a more modest 6.2% increase compared to the 21.4% increase for irrigation efficiency. This suggests that weather conditions have a proportionally larger impact on irrigation decisions than on output levels, consistent with farmers' ability to substitute other inputs for water when conditions are favorable.

Figure 4 shows the spatial distribution of average irrigation efficiency scores (with weather variables included). Highest efficiency scores appear in the extreme north and south portions of the aquifer and along the Texas–Oklahoma border, while lower scores are observed along the northern Platte River region and in parts of the southern High Plains.

More specifically, lower irrigation efficiency scores are observed along the northern part of the Platte River and in the North Central regions of Nebraska, as well as in parts of the southern High Plains region in Texas and New Mexico. Other counties with lower efficiency include Laramie County in Wyoming and Kit Carson County in Colorado.

Figure 5 illustrates how including weather variables affects measured irrigation efficiency across the HPA region. The spatial pattern reveals substantial heterogeneity in weather's impact.

The adjustment is particularly pronounced in southwestern Kansas and the Texas–Oklahoma Panhandle, where efficiency scores increase by 0.20–0.40 points (dark green counties). These counties face some of the harshest growing conditions in the region (combining low precipitation with high extreme degree days), making their irrigation levels appear excessive when compared to all counties, but justified when compared only to counties with similarly challenging weather.

In contrast, counties in Nebraska, particularly in the northern and eastern regions, show minimal weather adjustment (white or light green), with efficiency scores increasing by less than 0.10 points. These areas receive more rainfall and experience fewer extreme heat events, meaning the first-stage efficiency scores (without weather controls) were already comparing them to counties facing similar environmental conditions.

The pattern along central Nebraska, including counties near the Platte River, shows moderate adjustments (medium green, 0.10–0.20 points), reflecting the transitional climate zone between the wetter eastern plains and the semi-arid western regions. This geographic gradient indicates that weather heterogeneity is a significant driver of irrigation efficiency differences across the HPA region.

Specifically for the southern High Plains region in Texas, we observe significant heterogeneity in irrigation efficiency. While counties located in the south of the region (e.g., Dawson) show high technical efficiency levels, central areas such as Swisher and Deaf Smith counties display lower irrigation efficiency scores. The Llano Estacado, which intersects with the southern High Plains region, is economically important, as more than 20% of the total production of U.S. cotton is harvested from this region (Allen et al. 2008). Allen et al. (2008) also emphasize that the water pumping rate in the region historically exceeds the recharge rate of the aquifer.

Irrigation Productivity Change

To examine the role of weather in measuring productivity change over time, we calculate irrigation productivity using the Global Malmquist Index (equation 11) with pooled data. Table 5 presents results from the three-stage DEA that accounts for weather variables, while table 6 shows results without weather controls. We decompose productivity change into efficiency change and technical change using equation (13). Standard errors are obtained from a 1,000-replication Simar and Wilson (2000) bootstrap that resamples counties with replacement and re-computes the DEA frontiers for each census year in every draw.

Over the full period (1997–2017), counties increased irrigation productivity by 146%, meaning

they could generate the same crop sales per acre in 2017 using approximately 59% less irrigated land than required in 1997, holding constant fertilizer and chemical use and accounting for weather conditions.

Comparing tables 5 and 6 reveals that weather variables substantially affect measured productivity change. The model including weather (table 5) shows irrigation productivity growth of 146%, while the model excluding weather (table 6) shows only 120% growth. This 26 percentage point difference demonstrates that omitting weather variables understates the true productivity gains achieved by HPA counties.

The decomposition into efficiency change and technical change further illustrates weather's importance. Efficiency change is substantially higher when weather is included (96% vs. 34%), indicating that a significant portion of measured efficiency improvement captures counties' adaptation to local environmental conditions rather than purely technological or managerial advances. Conversely, measured technical change is lower with weather controls (34% vs. 77%), as the model without weather attributes some environmental variation to technological shifts.

The negative technical change observed in table 6 for drought years (1997–2002 and 2007–2012) illustrates the problem of omitting weather. The region experienced severe droughts in 2002 and 2012 (Hoerling et al. 2014). Without weather controls, these adverse conditions appear as contractions of the production frontier, lowering measured technical change. When weather is properly accounted for (table 5), the frontier adjustments more accurately reflect genuine technological change rather than environmental fluctuations.

Figure 6 shows the spatial distribution of irrigation productivity change from 1997–2017 (with weather variables included). Most HPA counties experienced productivity gains, particularly counties in Nebraska near the Platte River. A few counties in northwestern Nebraska (Garden, Cheyenne, and Sioux) and Oklahoma (Beaver and Ellis) showed productivity declines. Notably, some Nebraska counties with lower efficiency levels (figure 4) showed strong productivity gains, suggesting these counties are catching up to the regional frontier over time.

Overall, we notice that most counties in the HPA region experienced gains of irrigation pro-

ductivity to some degree. Among all counties in the Ogallala aquifer region, Beaver county and Ellis county, both in Oklahoma, experienced the most intensive loss of irrigation productivity from 1997 through 2017, while Platte county in Wyoming and Boone county in Nebraska exhibited the highest irrigation productivity change during the period.

Conclusions

This article examines how accounting for the effects of local weather conditions changes measures of both irrigation-oriented and output-oriented efficiency and productivity of irrigation practices in the High Plains Aquifer region. Using nonparametric methods including three-stage Data Envelopment Analysis and the Global Malmquist Index, we analyze county-level data from five agricultural census years (1997–2017) to compare results with and without weather variables (precipitation and extreme degree days). Since our analysis does not control for all production factors—including capital, labor, soil quality, managerial ability, local aquifer characteristics, and other variables that differ across counties and time—our measure of technical efficiency serves primarily as an indicator of regional heterogeneity rather than an assessment of input use and output production efficiency.

Our primary finding is that weather substantially reduces measured irrigation efficiency and productivity. Including weather contributions in the DEA model raises the average irrigation-oriented efficiency score by 21.4% (from 0.56 to 0.68) and the output-oriented score by 6.2% (from 0.81 to 0.86). The share of counties on the irrigation best-practice frontier doubles from 13.9% to 28.0%. These differences follow a distinct spatial pattern, with the largest adjustments (0.20–0.40 points) concentrated in the driest counties of southwestern Kansas and the Texas–Oklahoma Panhandle.

Weather variables also significantly affect measured productivity change. Over the full period (1997–2017), accounting for weather contributions yields a productivity growth measure of 146%, while the more traditional model considering only discretionary variables yields only 120%. The decomposition reveals that efficiency change is nearly three times higher when the effects of weather

are included (96% vs. 34%), while technical change is lower (34% vs. 77%). This pattern indicates that omitting weather attributes environmental adaptation to technological change.

The drought years of 2002 and 2012 particularly illustrate the importance of weather. Without weather controls, these adverse conditions appear as contractions of the production frontier, producing negative measured technical change. With weather accounted for, the model correctly identifies genuine technological progress separate from environmental fluctuations.

Together, these results demonstrate that weather heterogeneity is a critical determinant of both cross-sectional efficiency differences and temporal productivity changes in the High Plains Aquifer region. Omitting weather variables substantially understates both efficiency levels and productivity growth, while misattributing environmental adaptation to other sources of change.

Our analysis does not control for all production factors, including capital, labor, soil quality, managerial ability, local aquifer characteristics, and varying state regulations. Therefore, even after accounting for weather, the remaining variation should be interpreted as heterogeneity across counties rather than strict technical inefficiency. Future research incorporating these additional factors could further refine our understanding of irrigation practices in the region.

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Appendix

Table A1. Estimates from the Logistic Transformation Model in the Second Step of the DEA Approach for Output Efficiency (Equation 4)

Dependent variable:	$\text{logit}\left(1/\theta^{(1)*}\right)$
Precipitation	0.01
	(0.01)
Degree days (30°C)	−0.12**
	(0.05)
Intercept	0.38
	(0.84)
Num. obs.	518
R^2	0.61

Note: ***, **, and * indicate significance at 1%, 5%, and 10%, respectively. County and year fixed effects included.

Table A2. Estimates from the Logistic Transformation Model in the Second Step of the DEA Approach for Irrigation Efficiency (Equation 8)

Dependent variable:	$\text{logit}\left(\varphi^{(1)*}\right)$
Precipitation	0.02
	(0.01)
Degree days (30°C)	−0.13**
	(0.06)
Intercept	−1.35
	(1.04)
Num. obs.	518
R^2	0.61

Note: ***, **, and * indicate significance at 1%, 5%, and 10%, respectively. County and year fixed effects included.

Table 1. Descriptive Statistics for HPA Counties, 1997, 2002, 2007, 2012, and 2017

Variable	Units	Mean	SD	Min	Max
Crop sales (y)	Constant 2017 $/acre	312.11	125.31	28.13	775.07
Irrigation (w)	Fraction	0.47	0.23	0.03	0.96
Fertilizer (x_2)	Index	3.52	2.00	0.14	10.46
Chemicals (x_3)	Index	11.98	8.06	0.23	38.63
Precipitation (z_1)	Centimeters	35.90	14.06	9.46	81.71
Degree days 30°C (z_2)	24 hours	8.97	4.30	0.21	25.12
Number of observations	518				

Table 2. Mean and Standard Deviation of Variables by Year

		Crop sales ($/acre)	Irrigation	Fertilizer (index)	Chemicals (index)	Precip. (cm)	Degree days 30°C	Obs.
1997	Mean	256.04	0.44	3.04	7.49	38.57	6.67	109
	SD	86.38	0.21	1.59	4.34	9.87	2.58	
2002	Mean	294.53	0.50	3.39	7.44	26.21	11.00	101
	SD	112.30	0.23	1.89	4.71	8.22	3.18	
2007	Mean	325.05	0.49	3.56	11.06	48.31	6.74	104
	SD	104.50	0.23	2.06	6.20	15.55	2.38	
2012	Mean	278.05	0.49	3.49	16.18	23.65	13.73	104
	SD	115.51	0.23	2.12	8.60	6.16	4.48	
2017	Mean	412.96	0.44	4.15	18.06	42.62	6.78	100
	SD	143.01	0.22	2.19	8.89	9.95	2.92	

Table 3. Mean Technical Efficiency Scores (with Weather Variables) and Proportion of Technically Efficient Observations in HPA Counties (1997, 2002, 2007, 2012, and 2017)

Year	Obs.	Irrigation efficiency	Output efficiency	% Irr. efficient	% Output efficient
1997	109	0.47 (0.12)	0.82 (0.03)	19.3	15.6
2002	101	0.64 (0.05)	0.81 (0.03)	24.8	27.7
2007	104	0.69 (0.05)	0.88 (0.02)	31.7	26.9
2012	104	0.84 (0.02)	0.92 (0.01)	37.5	36.5
2017	100	0.76 (0.04)	0.89 (0.02)	32.0	35.0
Average	—	0.68 (0.04)	0.86 (0.02)	29.0	28.0

Note: Efficiency scores derived from the three-stage DEA approach that accounts for weather variables (precipitation and extreme degree days). Uses equation (6) for output-oriented efficiency and equation (10) for irrigation-oriented efficiency. Standard errors in parentheses.

Table 4. Mean Technical Efficiency Scores (without Weather Variables) and Proportion of Technically Efficient Observations in HPA Counties (1997, 2002, 2007, 2012, and 2017)

Year	Obs.	Irrigation efficiency	Output efficiency	% Irr. efficient	% Output efficient
1997	109	0.44 (0.09)	0.79	10.1	10.1
2002	101	0.57 (0.05)	0.77	11.9	11.9
2007	104	0.54 (0.04)	0.83	18.3	20.2
2012	104	0.73 (0.02)	0.84	16.3	16.3
2017	100	0.53 (0.05)	0.80	11.0	11.0
Overall mean	—	0.56 (0.03)	0.81	13.5	13.9

Note: Efficiency scores derived from the first-stage DEA approach without weather variables. Uses equation (3) for output-oriented efficiency and equation (7) for irrigation-oriented efficiency. Standard errors in parentheses.

Table 5. Average Irrigation Productivity Change, Technical Change, and Efficiency Change: Global Malmquist Index with Weather Variables Included

Period	Productivity change	Efficiency change	Technical change
1997–2002	1.78 (0.22)	1.60 (0.33)	1.11 (0.26)
2002–2007	0.79 (0.09)	1.08 (0.06)	0.73 (0.09)
2007–2012	1.55 (0.19)	1.26 (0.06)	1.22 (0.17)
2012–2017	1.13 (0.09)	0.87 (0.06)	1.29 (0.15)
1997–2017	2.46 (0.33)	1.96 (0.44)	1.34 (0.26)

Note: Values represent geometric mean of the variables by period. Only counties with observations for both 1997 and 2017 are included (27 counties dropped, 91 counties retained). Weather variables included. Bootstrap standard errors in parentheses.

Table 6. Average Irrigation Productivity Change, Technical Change, and Efficiency Change: Global Malmquist Index without Weather Variables

Period	Productivity change	Efficiency change	Technical change
1997–2002	1.15 (0.06)	1.49 (0.29)	0.77 (0.21)
2002–2007	1.10 (0.06)	0.91 (0.08)	1.20 (0.12)
2007–2012	0.73 (0.06)	1.46 (0.10)	0.50 (0.05)
2012–2017	2.38 (0.20)	0.68 (0.08)	3.50 (0.37)
1997–2017	2.20 (0.24)	1.34 (0.29)	1.77 (0.33)

Note: Values represent geometric mean of the variables by period. Only counties with observations for both 1997 and 2017 are included (27 counties dropped, 91 counties retained). Weather variables not included. Bootstrap standard errors in parentheses.

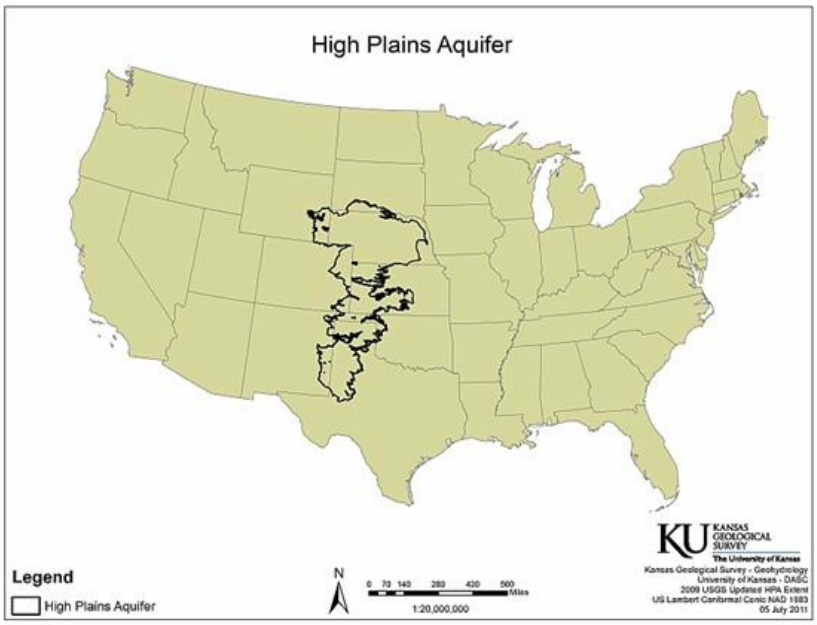


Figure 1. The Ogallala (or High Plains) Aquifer

Source: Kansas Geological Survey, The University of Kansas.

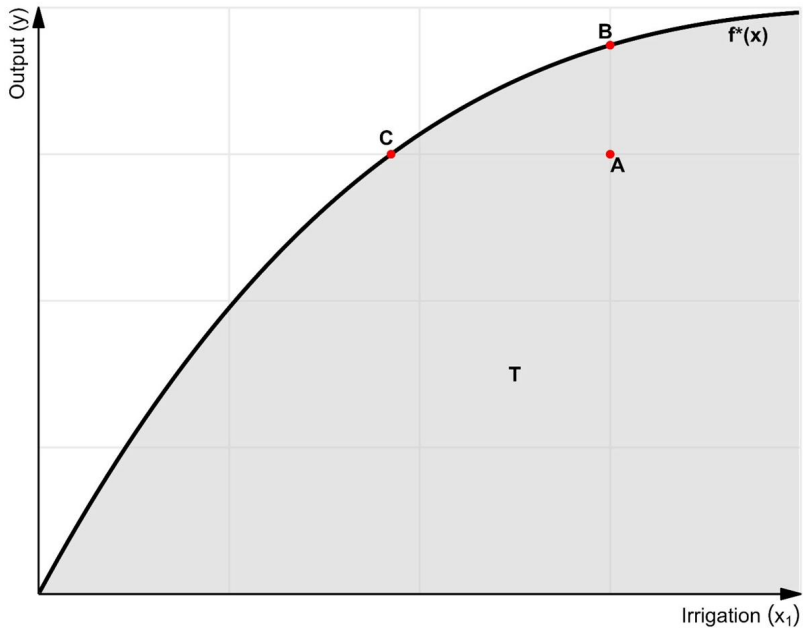


Figure 2. The feasible input-output set and the production frontier

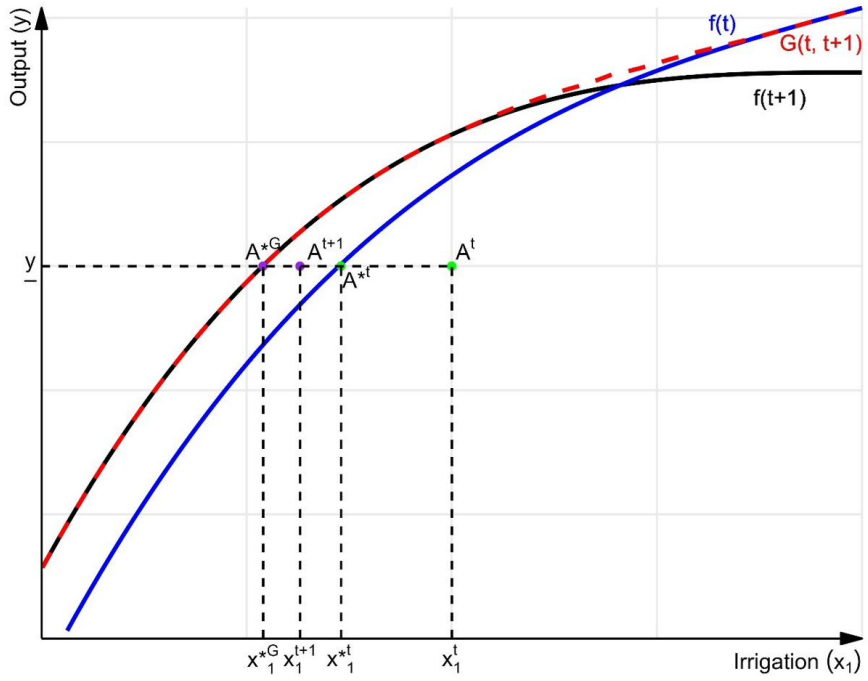


Figure 3. An illustration of the global frontier as envelope of the yearly frontiers, of global technical efficiency, of global technical change and of global irrigation productivity change from period t to period $t + 1$

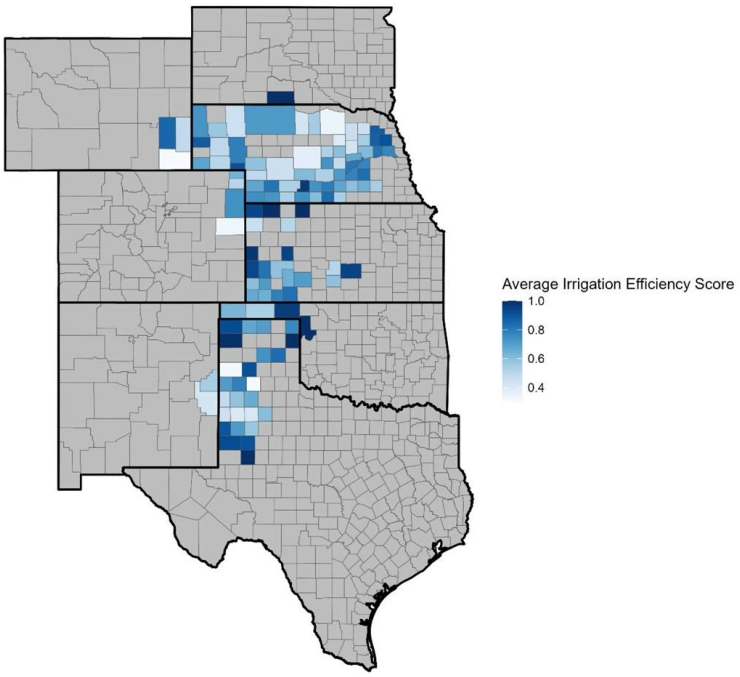


Figure 4. Distribution of overall mean irrigation efficiency scores (with weather variables) across HPA counties obtained from individual frontiers, agricultural census years 1997–2017

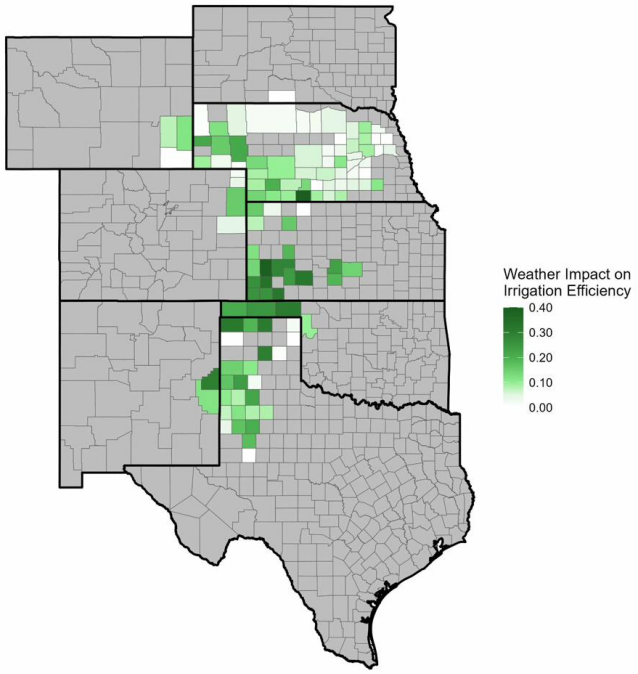


Figure 5. Difference in mean irrigation efficiency scores between models with and without weather variables across HPA counties, agricultural census years 1997–2017

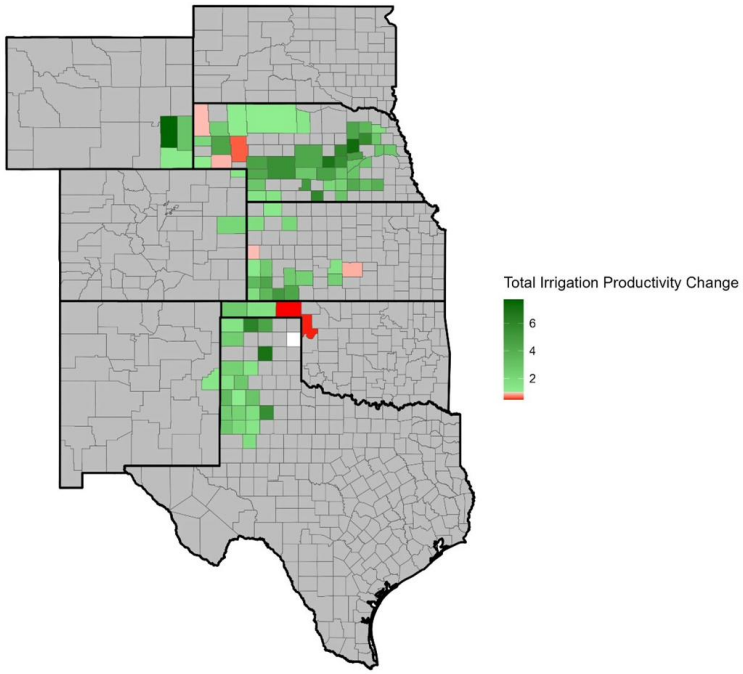


Figure 6. Spatial distribution of irrigation productivity change (with weather variables) across HPA counties, agricultural census years 1997–2017